Basic file handling

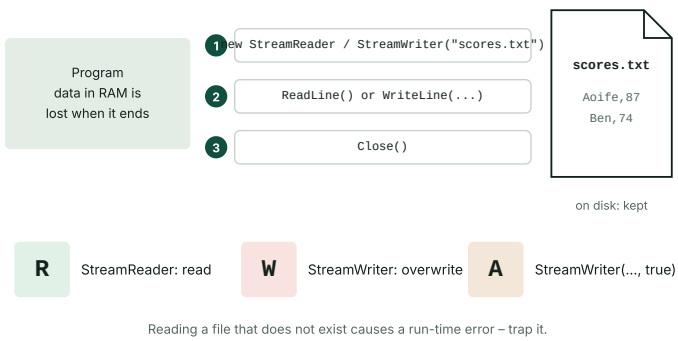
D20 FACT FILE

Specification: students should be able to

- Basic file handling

CCEA fact file: Programming Constructs 3

Key words: file, persistent storage, text file, open, read, write, append, close, file handle, mode, record



Opening, reading or writing, and closing a text file, with read, write and append modes

File handling

Data held in variables is lost when a program ends, so files provide persistent storage. To use a file a program opens it – for reading (StreamReader), writing (StreamWriter, which creates or overwrites) or appending (StreamWriter with true, which adds to the end) – reads or writes the data, then closes it. Text files often hold one record per line with fields separated by commas, which can be separated with Split(',').

T Key terms from the mark schemes

File handling
Opening, reading, writing, appending and closing files to store data permanently.
Credited: U4 2022 Q9, 2023 Q6

C C# example

Writing and appending

```
StreamWriter sw = new StreamWriter("scores.txt");        // write
sw.WriteLine("Aoife,87");
sw.WriteLine("Ben,74");
sw.Close();

StreamWriter sa = new StreamWriter("scores.txt", true);  // append
sa.WriteLine("Cara,91");
sa.Close();
```

Exam focus

2022 Q9(a): two file handling functions besides read and write (e.g. open, close, append, delete). 2023 Q6(b): storing orders for later reports. File handling is required in Unit 5 solutions.

Unit 4 2023 Q6	[10]	Candle shop billing algorithm; storing data in files
Unit 4 2022 Q9	[7]	File handling functions; three error types

? Check yourself

1. Explain why a program might store data in a file. [2]
2. Name and describe two file handling functions other than read and write. [4]
3. Describe the difference between write mode and append mode. [2]

Answers: 1. Data in variables is held in RAM, which is volatile, so it is lost when the program ends (1) / a file stores data permanently (persistent, non-volatile storage) so it can be loaded next time or shared with other programs/users (1) 2. Two functions, each named (1) + described (1), e.g.: open (1) – opens or creates a file ready for reading/writing in a chosen mode (1) / close (1) – closes the file so data is saved and the file is not left locked (1) / append (1) – adds data to the end of a file without deleting existing data (1) / delete (File.Delete) (1) – removes the file (1) / exists (File.Exists) (1) – checks whether a file exists before opening (1) 3. Write mode creates a new file or overwrites (deletes) all existing data in the file (1) / append mode adds new data to the end of the existing file, keeping existing data (1)